Notes: Jermann (RFS, 2020)
Negative Swap Spreads and Limited Arbitrage

Contents

1 Big picture

- **Question.** Why have long-maturity U.S. dollar swap spreads, especially 30-year swap spreads, been negative for long periods after 2008, even though textbook arbitrage arguments suggest swap spreads should not be negative?
- **Puzzle.** A fixed-for-floating swap can look cheap relative to Treasury bonds. An apparent arbitrage is to buy Treasuries and pay the lower fixed swap rate. If repo funding is cheap and LIBOR remains above repo, this generates positive cash flows. Yet negative spreads persisted.
- **Main explanation.** Arbitrage is limited because dealers face costs or constraints when holding government bonds. These costs prevent swap prices from being fully pinned down by Treasury bond prices.
- **Model structure.** Dealers trade short-term risk-free debt, long-term default-free debt, and fixed-for-floating swaps. Treasury prices are exogenous. Swap prices are endogenous and clear the swap market.
- **Key mechanism.** When long-term bond positions are costly, dealers hold fewer Treasuries and are less exposed to long-term interest-rate risk. They therefore require less compensation for receiving the fixed leg of a swap. The fixed swap rate declines, potentially below the Treasury yield.
- **Limiting intuition.** If bond-holding costs become extremely high, dealers hold no bonds. Then the expected swap rate is tied to expected LIBOR rather than to the long Treasury yield. Because the average TED spread is much smaller than the average term spread, the expected swap spread becomes negative.
- **Quantitative finding.** Moderate long-term bond holding costs can generate negative 30-year swap spreads close to the post-2008 data without needing large swap-demand imbalances.
- **Empirical implication.** Conditional on the short rate, term spreads should be negatively related to swap spreads. The regressions confirm this relation, especially for long maturities and post-crisis periods.
- **Economic takeaway.** Negative swap spreads are not necessarily arbitrage violations. They are equilibrium prices in a world where the institutions that would arbitrage swaps against bonds have costly balance-sheet capacity.

2 Data and measurement

2.1 Swap spread definition

A fixed-for-floating interest rate swap exchanges a fixed payment for a payment indexed to LIBOR. The **swap spread** is

$$\text{Swap spread}_t = y_t^{Sw} - \left(\exp(y_t^{LT}) - 1\right), \tag{1}$$

where y_t^{Sw} is the fixed swap rate and $\exp(y_t^{LT}) - 1$ is the yield on a default-free Treasury bond of the same maturity.

Interpretation: A negative spread means that the fixed swap rate is below the Treasury yield of the same maturity. This is counterintuitive because swaps have historically been viewed as private contracts whose fixed rate should exceed Treasury rates due to LIBOR credit/liquidity premia.

2.2 Securities in the model

The dealer can trade three securities:

- short-term default-free debt, interpreted as Treasury or repo funding;
- long-term default-free Treasury debt;
- fixed-for-floating swaps.

The short-term risk-free debt price is

$$q^{ST}(z) = \exp[-y^{ST}(z)]. \tag{2}$$

LIBOR debt has price

$$q^{LIB}(z) = \exp[-y^{LIB}(z)], \tag{3}$$

with

$$y^{LIB}(z) = y^{ST}(z) + \theta(z), \tag{4}$$

where $\theta(z)$ is the TED spread.

Interpretation: The TED spread is the short-term wedge between LIBOR and Treasury/repo funding. It is central because the floating leg of the swap is tied to LIBOR rather than to the Treasury short rate.

2.3 Long-term debt and term spread

Long-term default-free debt pays a coupon plus amortization. Its price is

$$q^{LT}(z) = \frac{c_{LT} + \lambda}{\exp[y^{LT}(z)] - 1 + \lambda}, \tag{5}$$

where $1/\lambda$ is average maturity. The long yield satisfies

$$y^{LT}(z) = y^{ST}(z) + \tau(z), \tag{6}$$

where $\tau(z)$ is the term spread.

Interpretation: The term spread is the risk premium and expected-rate component embedded in long Treasury yields. The paper's key insight is that this long-term bond risk premium need not be fully priced into swaps when bond arbitrage is costly.

2.4 Swap payoff and at-market swap rate

For the fixed-rate receiver, the swap payoff next period is

$$c_{Sw} - \left(\frac{1}{q^{LIB}(z)} - 1 \right) + (1 - \lambda) m', \tag{7}$$

where c_{Sw} is the fixed coupon, m' is the next-period mark-to-market value, and λ controls swap maturity. A newly issued at-market swap has $m = 0$. The coupon rate that makes $m = 0$ is the swap rate y^{Sw}.

Interpretation: The fixed swap rate is the price that clears the market for receiving fixed and paying floating. In a frictionless model it is tightly linked to bond yields. With costly bond positions, it is priced by a distorted dealer valuation.

2.5 Holding-cost measurement

The dealer faces quadratic holding costs for bond positions:

$$h(\alpha'_{ST}) = \frac{\kappa_{ST}}{2} (\alpha'_{ST})^2, \tag{8}$$

$$j(\alpha'_{LT}) = \frac{\kappa_{LT}}{2} (\alpha'_{LT})^2. \tag{9}$$

Interpretation: These costs are reduced-form balance-sheet frictions. They represent financing costs, regulatory capital/leverage costs, dealer risk-management limits, and other costs of holding Treasury positions needed to arbitrage swap spreads.

2.6 Empirical variables used in regressions

The empirical tests use monthly growth-rate regressions of swap spreads on:

- **TERM:** 30-year Treasury rate minus the 3-month Treasury rate;
- **TED:** the TED spread;
- **MBSD:** duration of mortgage-backed securities;
- **3MTB:** the 3-month Treasury rate;
- **VIX:** market volatility index.

Interpretation: TERM is the model's central empirical regressor. A higher term spread should be associated with a lower swap spread, especially at long maturities and after balance-sheet frictions became more binding.

3 Models and theoretical specifications

3.1 Dealer problem and distorted state prices

The dealer maximizes the lifetime utility of profits while trading bonds and swaps. The first-order conditions for short and long bonds are distorted by marginal holding costs:

$$q^{ST}(z) = \beta E\left[\frac{u_1(c')}{u_1(c)e^{\mu(z')}}\right] - h_1(\alpha'_{ST}), \tag{10}$$

$$q^{LT}(z) = \beta E\left[\frac{u_1(c')}{u_1(c)e^{\mu(z')}}\left(c_{LT} + \lambda + (1-\lambda)q^{LT}(z')\right)\right] - j_1(\alpha'_{LT}). \tag{11}$$

The swap price satisfies

$$m = \beta E\left[\frac{u_1(c')}{u_1(c)e^{\mu(z')}}\left(c_{Sw} - \left(\frac{1}{q^{LIB}(z)} - 1\right) + (1-\lambda)m'\right)\right]. \tag{12}$$

Bond holding costs put wedges in the bond Euler equations. The swap Euler equation is not directly penalized in the baseline model. Therefore the state prices that value swaps are not forced to be exactly consistent with the state prices implied by Treasury bonds.

3.2 No-arbitrage benchmark

If holding costs are zero, the dealer can freely trade bonds and swaps. In that case the swap is priced by the same valuation system as Treasuries. The swap spread is positive because the floating leg is tied to LIBOR, and LIBOR is above the short-term Treasury/repo rate through the TED spread.

This benchmark is the textbook case: a swap and a bond are linked by arbitrage. Negative spreads are hard to reconcile with this world.

3.3 Limited-arbitrage mechanism

When $\kappa_{LT} > 0$, holding long-term bonds is costly. Dealers therefore reduce long Treasury positions. Lower long-bond exposure implies less compensation for long-term interest-rate risk in the swap. The fixed swap rate falls.

The paper shows that the swap spread depends on:

- the TED spread;
- the term spread;
- dealer valuations over future states;
- the size of dealer bond positions and marginal holding costs.

The swap spread can become negative without assuming a large exogenous demand shock in the swap market. The friction by itself changes how swap cash flows are priced.

3.4 Very strong frictions and the negative-spread limit

In the limiting case of very high bond-holding costs, dealers hold no bonds. With no inflation uncertainty, the expected swap rate equals expected LIBOR:

$$E[y_t^{Sw}] = E[y_t^{LIB}] = E[y_t^{ST}] + E[\theta_t]. \tag{13}$$

The expected Treasury yield is

$$E[y_t^{LT}] = E[y_t^{ST}] + E[\tau_t]. \tag{14}$$

Therefore the expected swap spread is

$$E[y_t^{Sw} - y_t^{LT}] = E[\theta_t] - E[\tau_t]. \tag{15}$$

Historically, the paper reports average annualized values of approximately $E(\theta_t) = 0.6\%$ and $E(\tau_t) = 1.7\%$, implying

$$E[y_t^{Sw} - y_t^{LT}] \approx -1.1\%. \tag{16}$$

The limiting case provides the paper's clean intuition. If swaps no longer inherit the long-bond risk premium, then their fixed rate is tied to LIBOR. Since the term spread is much larger than the TED spread on average, the swap spread is naturally negative.

3.5 Demand effects and swap holding costs

The paper extends the baseline with demand for fixed-receiver swaps and direct holding costs for swaps. Demand for receiving fixed lowers the fixed swap rate. Swap holding costs raise or lower swap rates depending on the sign of the dealer's swap position.

Demand and swap-specific costs can move spreads, but the quantitative results suggest that the core negative-spread result does not require these channels to be large.

3.6 Leverage constraint

The paper also studies a leverage-type constraint on dealer positions. A tighter constraint increases the implicit marginal cost of holding long-term bonds, much like a higher κ_{LT}.

The leverage-constraint extension ties the reduced-form holding cost to post-crisis bank balance-sheet regulation. It supports the interpretation that capital and leverage rules can make Treasury-swap arbitrage costly.

3.7 Regression specification

The empirical regressions relate monthly changes in swap spreads to changes in macro-financial variables:

$$\Delta SS_t^n = a_n + b_n \Delta TERM_t + c_n \Delta TED_t + d_n \Delta MBSD_t + e_n \Delta 3MTB_t + f_n \Delta VIX_t + \varepsilon_t^n, \tag{17}$$

where n indexes swap maturity.

The model predicts $b_n < 0$, especially for long maturities. A higher term spread makes long Treasury exposure more valuable relative to swaps, but if arbitrage is constrained, the swap spread falls instead of fully absorbing the term premium.

4 Identification and empirical design

4.1 What identifies the mechanism in the model

- Treasury yields and the TED spread are taken as exogenous state variables.
- Swap rates are determined endogenously by dealers subject to holding frictions.
- Comparing the frictionless case to higher κ_{LT} isolates the effect of limited arbitrage.
- The limiting case with no bond holdings provides an analytical explanation for why spreads become negative.

The model is not estimating a structural causal parameter from a natural experiment. It shows that a plausible equilibrium mechanism can rationalize observed negative spreads.

4.2 Why the 2008 and 2015 episodes matter

- Long-maturity swap spreads became negative after the financial crisis.
- The sharp decline in 2015 coincides with the period when large U.S. banks began publicly disclosing Supplementary Leverage Ratios.
- These facts motivate balance-sheet and regulatory frictions as the relevant limit to arbitrage.

The timing does not prove regulation is the only driver, but it is consistent with the model's balance-sheet capacity interpretation.

4.3 Why term-spread regressions are the empirical test

The model implies that, conditional on the short rate, a higher term spread should lower the swap spread. This relationship is not necessarily strong unconditionally because many state variables move together. The regressions therefore control for short rates, TED spreads, mortgage duration, and volatility.

The negative coefficient on TERM is the main reduced-form evidence for the mechanism. It says that when the long Treasury term premium rises, swaps do not fully price that premium because arbitrage is limited.

4.4 What the paper cannot fully prove

- The model takes Treasury yields as exogenous and prices swaps only.
- Holding costs are reduced-form and not directly measured for every dealer.
- The model does not explicitly estimate dealer balance sheets, repo haircuts, or bank-level regulatory constraints.
- The regression evidence is consistent with the model but not a complete causal identification of regulation.
- Demand effects, collateral terms, funding-market stress, and institutional hedging flows could also contribute to swap spread movements.

The paper's contribution is to show that negative swap spreads are theoretically natural under limited arbitrage and quantitatively plausible with moderate frictions.

5 Tables and figures

5.1 Figure 1: Swap spreads

Read:

- The figure plots fixed swap rates minus Treasury yields for 2-, 5-, 10-, 20-, and 30-year maturities from 1998 to 2018.
- Pre-2008 swap spreads are generally positive, though they vary by maturity.
- Around the 2008 crisis, longer-maturity spreads fall sharply.
- The 30-year swap spread remains negative for years and falls particularly sharply around 2015.
- Shorter maturities are less persistently negative, showing that the puzzle is strongest at long maturities.

Interpretation: The figure establishes the core fact. Long-maturity swaps began trading with fixed rates below same-maturity Treasury yields. In a frictionless arbitrage model, this should be difficult to sustain. The persistence of the pattern implies that the impediment to arbitrage must be persistent, not merely a temporary pricing error.

Economic message: The visual pattern motivates a balance-sheet-friction explanation: the trade that would close the spread requires holding Treasuries, financing them, and bearing regulatory/capital costs. Those costs matter most for long maturities.

What not to over-read: The figure by itself does not identify the cause of negative spreads. It shows the timing and maturity pattern, but the mechanism comes from the model and regression evidence.

5.2 Table 1: Model parameters

Read:

- The calibrated short-rate level is 0.01156.
- The term-spread level is 0.00429.
- The TED spread level is 0.00158.
- Risk aversion is set to $\gamma = 2$ and discount elasticity to $\nu = 1$.
- The long-term bond and swap maturity parameter is $1/\lambda = 120$, corresponding to a long average maturity.

Interpretation: The parameterization is designed to match key interest-rate facts while keeping the model parsimonious. The maturity parameter is especially important because the paper is focused on 30-year swap spreads.

Economic message: The model uses observed interest-rate moments to discipline the benchmark. The frictions then determine whether the same interest-rate environment produces positive or negative swap spreads.

What not to over-read: Table 1 is not a full structural estimation of all dealer costs. The main friction parameter κ_{LT} is varied in later tables to examine how costly bond arbitrage affects spreads.

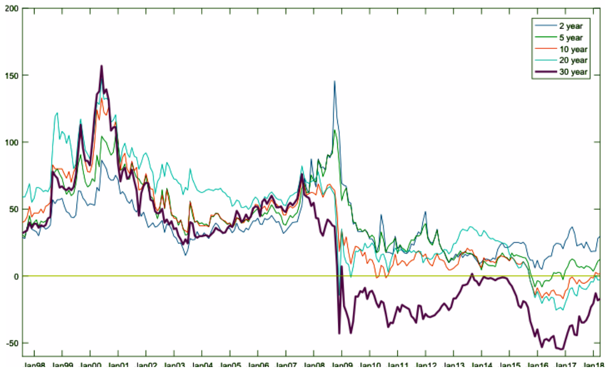


Table 1
Model parameters

Symbol	Parameter	Value
$\bar{y}_{ST}$	Short rate level	0.01156
$\bar{\tau}$	Term spread level	0.00429
$\bar{\mu}$	Inflation level	0.00938
$\bar{\theta}$	TED spread level	0.00158
γ	Risk aversion	2
ν	Discount elasticity	1
$1/\lambda$	Maturity of long-term debt and swap	120

5.3 Table 2: Swap spreads in the data and the model

Read:

- In the data, the 30-year swap spread mean is 57 bps from July 1997 to September 2008, but -18 bps from October 2008 to October 2015.
- In the model, a low friction $\kappa_{LT} = 0.0001$ produces a positive mean spread of 62 bps.
- The benchmark friction $\kappa_{LT} = 0.0032$ produces a mean spread of -18 bps, matching the post-2008 average.
- A high friction $\kappa_{LT} = 0.01$ produces a much more negative mean spread of -54 bps.
- The implied marginal cost of long-term bond holding rises with κ_{LT}, reaching 102 bps in the benchmark case and 119 bps in the high-friction case.

Interpretation: This is the central quantitative table. It shows that increasing long-term bond-holding costs can move the model from a positive-spread world to a negative-spread world. The benchmark calibration is chosen to match the post-2008 level, while alternative rows show that the mechanism is robust to risk aversion, discount elasticity, TED dynamics, inflation dynamics, and short-term bond holding costs.

Economic message: A moderate cost of holding long Treasuries can rationalize the observed negative spread. The table makes the "limited arbitrage" explanation quantitative: the apparent arbitrage persists because the balance-sheet cost of executing it is material.

What not to over-read: Matching the post-2008 mean does not prove that $\kappa_{LT} = 0.0032$ is the true underlying cost. The table shows plausibility and magnitudes, not a unique estimate of regulatory cost.

Table 2
Swap spreads in the data and the model

	E(30Y SS)	SD(30Y SS)	κ_{LT}E(α_{LT})	E(TED)
Data				
$Jul.1997 - Sep.2008$	57	27		58
$Oct.2008 - Oct.2015$	−18	12		35
Model				
$\kappa_{LT}=0.0001$	62	8	21	63
κ_{LT}=**0.0032**	**−18**	**61**	**102**	**63**
$\kappa_{LT}=0.01$	−54	89	119	63
Post-October 2008 TED level and $\kappa_{LT}=0.0014$	−18	41	80	35
Higher risk aversion, $\gamma=4$	−36	66	119	63
recalibrated, $\kappa_{LT}=0.00195$	−18	54	106	63
Lower discount elast., $\nu=0.8$	−26	67	106	63
recalibrated, $\kappa_{LT}=0.00255$	−18	61	102	63
Short term debt cost, $\kappa_{ST}=0.0032$	−15	82	110	63
Constant TED	−24	68	102	63
Constant Inflation	−27	64	115	63

Units are annualized basis points (i.e., multiplied by 40,000). Unless indicated otherwise, the cost parameter for

Table 3
Demand effects and swap costs

	Swap rate	Demand sensitivity	LT bond position	Swap cost impact	Marginal cost for swap
Benchmark ($\kappa_{LT}=0.0032$)					
$d=0$	6.08		1.05		
$d=.2$	5.92	−.99	1.14		
$d=.2, \kappa_{Sw}=.005$	5.52			−.4056	.4000
Low friction ($\kappa_{LT}=0.0001$)					
$d=0$	7.00		3.52		
$d=.2$	6.99	−.14	3.72		
$d=.2, \kappa_{Sw}=.005$	6.58			−.4056	.4000
High friction ($\kappa_{LT}=0.01$)					
$d=0$	5.67		.46		
$d=.2$	5.40	−1.28	.51		
$d=.2, \kappa_{Sw}=.005$	5.00			−.4028	.4000

Units for rates and marginal costs are annualized percentage points (i.e., multiplied by 400). The demand

5.4 Table 3: Demand effects and swap costs

Read:

- The table studies demand for receiving fixed swaps and direct swap holding costs.
- In the benchmark case, increasing demand from $d = 0$ to $d = 0.2$ reduces the swap rate from 6.08 to 5.92.
- Adding swap costs can reduce the swap rate further, for example to 5.52 in the benchmark case.
- The long-term bond position increases when demand rises because dealers hedge swap exposure with long bonds.
- The reported demand sensitivity varies across friction regimes but remains meaningful.

Interpretation: Demand effects can move swap rates. If end users want to receive fixed, dealers are on the other side and have exposure that can be partly hedged by long bonds. However, the table also shows that demand effects require meaningful positions and are not the paper's main explanation for post-2008 negative spreads.

Economic message: Swap-demand imbalances matter, but they operate through dealer balance sheets. If the balance sheet is costly, even moderate demand or hedging pressure can affect swap rates.

What not to over-read: Do not read Table 3 as saying demand pressure is irrelevant. The more precise message is that the model does not need a large demand imbalance to generate negative spreads; demand is a complementary channel.

5.5 Table 4: Model with leverage constraint

Read:

- With no leverage constraint, the model produces a positive mean 30-year swap spread of 62 bps.
- As the leverage multiplier ξ falls, the constraint tightens.
- With $\xi = 10$, the mean spread falls to 19 bps and the spread is negative 20% of the time.
- With $\xi = 5$, the mean spread becomes -25 bps and the spread is negative 66% of the time.
- The constraint binds mostly through long-term bond positions, not short-term positions.

Interpretation: This table maps the reduced-form holding-cost idea into a leverage/capital constraint. A tight leverage constraint acts like a cost on Treasury arbitrage positions. It can make swap spreads negative because it prevents dealers from taking the long Treasury positions needed to close the spread.

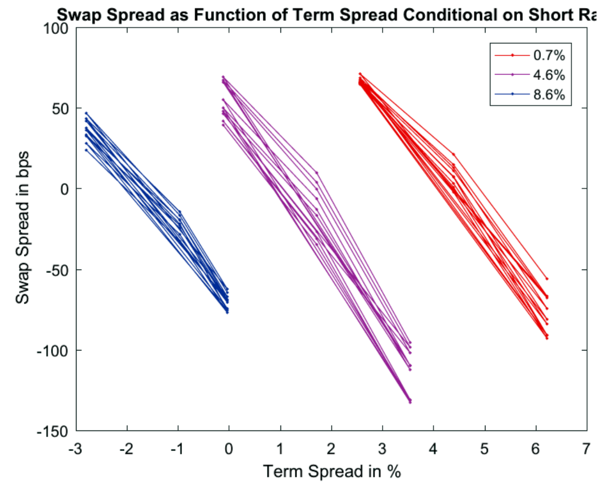
Economic message: Regulation and balance-sheet capacity can directly affect derivative prices. Even if swaps themselves are off-balance-sheet or collateralized, arbitraging swap spreads requires balance-sheet-intensive Treasury positions.

What not to over-read: The table does not say the exact SLR requirement alone mechanically explains all negative spreads. The paper notes that a very tight constraint is needed to produce negative average spreads from the leverage constraint alone.

Table 4
Model with leverage constraint

Leverage multiplier	E(30Y SS)	SD(30Y SS)	freq(SS$<$0)	freq($\eta>0$)	freq($\alpha_{LT}>0 \mid \eta>0$)
$\xi=\infty$	62	8	0	0	
$\xi=20$	48	9	.001	.05	1
$\xi=10$	19	24	.20	.14	1
$\xi=5$	−25	53	.66	.29	.90

This is the benchmark model version with very low long-term bond cost $\kappa_{LT}=0.0001$, augmented to include the leverage constraint. α_{LT} is the long-term bond position, η is the multiplier on the constraint.



5.6 Figure 2: Swap spread as a function of term spread conditional on short rate

Read:

- The figure plots model-implied swap spreads against the term spread.
- Each color corresponds to a different short-rate level: 0.7%, 4.6%, and 8.6%.
- Within each short-rate group, the relation between term spread and swap spread is strongly negative.
- When the term spread is high, the swap spread is lower, often negative.
- The downward slope is the model's main empirical prediction.

Interpretation: The figure shows the conditional relation that the regressions later test. The logic is that the term spread represents compensation embedded in long Treasury yields. When dealers cannot freely arbitrage long bonds against swaps, the swap rate does not fully incorporate that compensation; therefore the swap spread falls as the term spread rises.

Economic message: The figure explains why negative swap spreads should be especially plausible when long-bond term premia are high and arbitrage capacity is limited.

What not to over-read: The figure is a model implication, not a direct scatter plot of the raw data. The empirical support comes from Tables 5-7.

5.7 Table 5: Swap spread regressions, July 1997-May 2018

Read:

- TERM has a negative coefficient at all maturities.
- The negative relation is strongest for the 30-year maturity, with a coefficient of -0.205 and statistical significance at the 1% level.
- The TED spread has positive effects at shorter maturities but little effect at 30 years.
- Mortgage-backed-security duration is positively related to swap spreads, consistent with hedging-demand effects.

- The adjusted R^2 for the 30-year regression is 0.18, and excluding TERM lowers it to 0.08.

Interpretation: The table supports the model's prediction that term spreads and swap spreads are negatively related. The effect is strongest for long maturities, where the balance-sheet cost of Treasury positions should matter most.

Economic message: The regression evidence says that the term premium embedded in Treasuries is not fully passed into swaps. This is exactly what limited arbitrage predicts.

What not to over-read: The regression is not a natural experiment. It is evidence of a robust conditional relationship consistent with the model, not definitive proof that dealer regulation alone caused negative spreads.

Table 5
Swap spread regressions, July 1997–May 2018

Regressor \ Swap maturity	2	5	10	20	30
TERM	−0.036	−0.044	−0.157***	−0.128***	−0.205***
TED	0.220***	0.089***	0.025	0.012	0.007
MBSD	0.030**	0.043***	0.088***	0.058***	0.104***
3MTB	0.012	0.000	0.083*	−0.096	−0.100
VIX	0.004***	0.006***	0.002	−0.000	−0.000
Adj. R-squared	0.50	0.28	0.22	0.04	0.18
Adj. R-squared without TERM	0.49	0.27	0.12	0.00	0.08

All variables are in monthly growth rates. TERM stands for the difference between the 30-year Treasury rate minus the 3-month rate; TED is the TED spread; 3MTB is the 3-month Treasury rate; MBSD is the duration of mortgage-backed securities; and VIX the market volatility index. * p <.1; ** p <.05; *** p <.01. The regressions are estimated by OLS with a constant, with Newey-West standard errors.

Table 6
Swap spread regressions, October 2008–May 2018

Regressor \ Swap maturity	2	5	10	20	30
TERM	−0.045**	−0.008	−0.116***	−0.115*	−0.187**
TED	0.337***	0.179***	−0.102*	−0.285***	−0.253**
MBSD	0.026***	0.015	0.067***	0.065**	0.100***
3MTB	0.151*	0.154	−0.011	0.022	0.036
VIX	0.003	0.003**	0.002**	0.000	−0.001
Adj. R-squared	0.56	0.38	0.28	0.32	0.31
Adj. R-squared without TERM	0.55	0.39	0.15	0.27	0.20

All variables are in monthly growth rates. TERM stands for the difference between the 30-year Treasury rate minus the 3-month rate; TED is the TED spread; 3MTB is the 3-month Treasury rate; MBSD is the duration of mortgage-backed securities; and VIX is the market volatility index. * p <.1; ** p <.05; *** p <.01. The regressions are estimated by OLS with a constant, with Newey-West standard errors.

5.8 Table 6: Swap spread regressions, October 2008-May 2018

Read:

- In the post-crisis sample, TERM remains negative for 2-, 10-, 20-, and 30-year swap spreads.
- For the 30-year maturity, the TERM coefficient is -0.187 and statistically significant.
- TED becomes negative at longer maturities in this sample, unlike in the full sample.
- MBSD remains positive and significant for 2-, 10-, 20-, and 30-year maturities.
- TERM contributes meaningfully to the adjusted R^2 for long maturities.

Interpretation: The post-crisis table is important because negative swap spreads are mainly a post-2008 phenomenon. The persistence of the negative TERM coefficient strengthens the limited-arbitrage interpretation.

Economic message: After the crisis, balance-sheet frictions became more salient. The data show that long-maturity swap spreads move more negatively with term spreads in this environment.

What not to over-read: The post-crisis sample still contains multiple shocks: regulation, funding stress, QE, changes in mortgage hedging, and changes in dealer balance sheets. The regression isolates correlations, not all structural channels.

5.9 Table 7: Swap spread regressions, January 2015-May 2018

Read:

- The 2015-2018 period focuses on the years after public disclosure of Supplementary Leverage Ratios.

- TERM is strongly negative for 10-, 20-, and 30-year maturities.
- The 30-year TERM coefficient is -0.455 and statistically significant at the 1% level.
- The 3-month Treasury rate also has a large negative coefficient for long maturities.
- The explanatory power is limited for some shorter maturities, but the 30-year regression has adjusted $R^2 = 0.26$.

Interpretation: This table connects the empirical pattern most directly to regulation-related balance-sheet constraints. The strongest negative relationship appears in the period when bank leverage disclosure and related balance-sheet considerations were especially prominent.

Economic message: The long end of the swap curve behaves as the model predicts when dealer balance-sheet capacity is tight: higher term spreads are associated with much lower swap spreads.

What not to over-read: The sample is short, so coefficients should be interpreted cautiously. The table is suggestive evidence that the post-2015 decline in long swap spreads is consistent with the model, not a complete event-study proof.

Table 7
Swap spread regressions, January 2015–May 2018

Regressor \ Swap maturity	2	5	10	20	30
TERM	0.030	−0.052	−0.169***	−0.238***	−0.455***
TED	0.310***	0.146***	−0.076*	0.050	0.099*
MBSD	−0.003	0.013	0.075***	0.111***	0.201***
3MTB	0.151	−0.020	−0.214***	−0.326***	−0.487**
VIX	−0.001	0.001	0.001	0.001	0.000
Adj. R-squared	0.36	0.08	0.09	0.10	0.26
Adj. R-squared without TERM	0.38	0.10	0.01	0.00	0.00

All variables are in monthly growth rates. TERM stands for the difference between the 30-year Treasury rate minus the 3-month rate; TED is the TED spread; 3MTB is the 3-month Treasury rate; MBSD is the duration of mortgage-backed securities; and VIX is the market volatility index. * $p < .1$; ** $p < .05$; *** $p < .01$. The regressions

Table 7: Swap spread regressions, January 2015-May 2018.

6 Short summary

- Negative long-maturity swap spreads became persistent after 2008, especially at the 30-year maturity.
- In a frictionless no-arbitrage framework, this is hard to explain because buying Treasuries and paying fixed in swaps appears to be an arbitrage.
- Jermann builds a model in which dealers price swaps but face costs or constraints when holding bonds.
- Bond holding costs distort the bond Euler equations and prevent swap pricing from being fully tied to Treasury pricing.
- As long-term bond holding costs rise, dealers hold fewer long bonds, require less compensation for swap fixed-rate exposure, and swap rates fall.
- In the limiting case of very high bond costs, expected swap spreads equal the average TED spread minus the average term spread, which is negative historically.
- The calibrated model can generate the post-2008 30-year swap spread average with a moderate long-bond holding cost.
- Demand effects and swap-specific costs can matter, but they are not necessary for the main negative-spread result.
- A leverage constraint works similarly to long-bond holding costs and helps connect the model to post-crisis regulation.

- Empirical regressions show that term spreads are negatively related to swap spreads, especially at long maturities and after 2008/2015.
- The paper's main contribution is to make negative swap spreads an equilibrium outcome of limited arbitrage rather than a simple arbitrage anomaly.

7 Conclusion

7.1 Core conclusion

Jermann shows that negative swap spreads are consistent with equilibrium pricing once the arbitrage between swaps and Treasuries is performed by constrained intermediaries. The key friction is not that swaps are mispriced in a mechanical sense, but that the institutions capable of exploiting the apparent arbitrage face balance-sheet costs when holding the Treasury leg of the trade.

7.2 Economic intuition

In a frictionless world, swaps and Treasury bonds are tightly linked. A fixed swap rate below the Treasury yield creates a trade that seems to generate positive cash flows. But the trade is not free: it requires financing long Treasury positions, using dealer balance sheet, bearing interest-rate and funding risk, and satisfying regulatory constraints. If these costs are high enough, the swap rate no longer needs to fully reflect the long Treasury term premium. The swap rate instead moves closer to expected LIBOR. Since the average TED spread is lower than the average term spread, negative swap spreads become natural.

7.3 Why the result matters

The paper matters because swap spreads are often treated as near-arbitrage pricing relations. The evidence since 2008 shows that even very liquid, plain-vanilla derivative markets can be strongly affected by intermediary constraints. This links derivative pricing to the broader intermediary asset-pricing literature: prices are shaped not only by cash flows, but also by who can hold the offsetting positions and how costly those positions are.

7.4 What the paper teaches about regulation

The paper does not claim that one specific regulation mechanically caused all negative swap spreads. Instead, it shows that leverage and balance-sheet costs have the right qualitative and quantitative effect. Post-crisis regulation, supplementary leverage ratios, repo balance-sheet costs, and capital charges can all make the Treasury leg of the swap-spread arbitrage expensive. This helps explain why the decline in long swap spreads persisted rather than being immediately arbitraged away.

7.5 Limitations and open questions

Important mechanisms remain outside the model. Treasury yields are exogenous. Dealer heterogeneity is not modeled directly. Repo-market frictions, collateral terms, central clearing, regulatory implementation details, and client hedging flows are simplified or represented in reduced form. The model also focuses on swap spreads rather than jointly pricing Treasuries, swaps, repo, and bank balance sheets. These are natural directions for extensions.

7.6 Takeaway

The main takeaway is simple: negative swap spreads are a balance-sheet-capacity phenomenon. When arbitrage requires scarce intermediary balance sheet, derivative prices can depart from textbook no-arbitrage values for long periods. In this setting, “arbitrage” exists only after paying the cost of holding and financing the hedging asset, and that cost can be large enough to make negative swap spreads an equilibrium outcome.